Set Theory Through Maps: From Characteristic Functions to Cantor's Theorem

N-20220131

A set can be understood by the ways other sets map into it and by the ways it maps into others. This perspective is especially useful when a proof seems to depend on a mysterious encoding of mathematical objects as sets. Products, subsets, elements and functions can all be recovered from a small collection of mapping principles. Cantor's theorem then becomes a direct consequence of how evaluation and negation interact.[1]

§1.1 An element is a map from a singleton

In familiar set theory, an element $x \in X$ is commonly regarded as a primitive relation. Instead fix a one-element set $1 = \{*\}$. Every element $x \in X$ determines a unique map

$$\widehat{x} : 1 \longrightarrow X, \qquad \widehat{x}(*) = x.$$

Conversely every such map determines one element. Thus

$$\boxed{\text{elements of } X \;\longleftrightarrow\; \mathrm{Hom}(1, X).}$$

This is a small example of a useful change in language: instead of specifying what an object is made of, one specifies maps that probe it.

For example, a loop in a space X is represented by a map $S^1 \to X$, while a sequence of elements of X is a map $\mathbb{N} \to X$. An element is simply a probe with the smallest nonempty domain.

§1.2 The Universal Property of Products

The cartesian product $X \times Y$ comes with projection maps

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y.$$

Its decisive property is that, for every set T and maps $f : T \to X$, $g : T \to Y$, there is a unique map $\langle f, g\rangle : T \to X \times Y$ satisfying

$$\pi_X \circ \langle f, g\rangle = f, \qquad \pi_Y \circ \langle f, g\rangle = g.$$

Taking $T = 1$ recovers the usual description of a pair as two elements. But the universal property applies equally well to pairs of functions, pairs of families or pairs of continuously varying parameters in a different category.

It also proves uniqueness of products up to a unique compatible isomorphism. If P and Q both satisfy the same property, the property constructs maps $P \to Q$ and $Q \to P$; their composites must be the respective identities by uniqueness. Thus a product is characterized by what it does, independent of an arbitrary encoding of ordered pairs.

[1] Tom Leinster, *Rethinking Set Theory* (2014), with his supplementary ETCS axioms.

§1.3 Function spaces and currying

An element of Y^X is itself a function $X \to Y$. That observation can be made universal. Given a map

$$q : T \times X \longrightarrow Y,$$

hold the T-coordinate fixed and regard the result as a function of X. We obtain a map

$$\widehat{q} : T \longrightarrow Y^X, \qquad \widehat{q}(t)(x) = q(t, x).$$

Conversely $h : T \to Y^X$ determines $(t, x) \mapsto h(t)(x)$. The constructions are inverse, and they commute with maps between parameter sets:

$$\boxed{\mathrm{Hom}(T \times X, Y) \cong \mathrm{Hom}(T, Y^X).}$$

This is the universal property of an *exponential object*. Just as projections characterize products, evaluation characterizes function spaces. The familiar act of treating a two-variable function as a family of one-variable functions is the map called *currying*.

For a very small example, take $T = \{a, b\}$ and $X = Y = \{0, 1\}$. Define q by the table

$q(t,x)$	$x=0$	$x=1$
$t=a$	0	1
$t=b$	1	0

.

Then $\widehat{q}(a)$ is the identity function on $\{0, 1\}$ and $\widehat{q}(b)$ is Boolean negation. The very same four values can be read row-by-row as two functions, or all at once as one function of two inputs.

The importance of this property is its independence from the chosen description of a function. A category can have exponential objects even when its arrows are continuous maps, group actions, or something less familiar. In each case one must check whether the appropriate function object exists. The evaluation map gives a convenient way to formulate that question.

§1.4 Equations and fibers as universal constructions

There is a similarly economical description of the solution set of an equation. Given maps $f, g : X \to Y$, their *equalizer* is a map $e : E \to X$ such that $fe = ge$, universal among maps on which f and g agree. In ordinary sets,

$$E = \{x \in X : f(x) = g(x)\}.$$

If a map $h : T \to X$ satisfies $fh = gh$, there is exactly one $\bar{h} : T \to E$ with $e\bar{h} = h$. This last sentence still makes sense when the elements of E have never been specified.

A fiber is a special case. Fix $y : 1 \to Y$ and consider $f : X \to Y$. Then $f^{-1}(y)$ sits in a pullback square

$$\begin{array}{ccc} f^{-1}(y) & \xhookrightarrow{\ i\ } & X \\ \downarrow & & \downarrow{\scriptstyle f} \\ 1 & \xrightarrow[\ y\ ]{} & Y. \end{array}$$

One can recognize this square by a factorization test: $h : T \to X$ factors through the fiber exactly when fh is the constant map with value y.

As a calculation, let $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ send each integer to its residue. The fiber over $\overline{1}$ consists of $1, 4, 7, \ldots$ and their negative counterparts, namely $1 + 3\mathbb{Z}$. A map $h : T \to \mathbb{Z}$ factors through this fiber precisely when every value of h is congruent to 1 modulo 3.

Finite products and equalizers give finite limits, including all finite pullbacks. The everyday constructions of pairs, solutions to equations, and inverse images therefore grow from a rather short list of demands on arrows. Later, these same diagrams can be interpreted in categories whose objects have topology, geometry, or varying local data.

§1.5 Subsets as Binary Predicates

Let $2 = \{0, 1\}$. A subset $A \subseteq X$ determines its characteristic function

$$\chi_A : X \to 2, \qquad \chi_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Conversely, a function $\chi : X \to 2$ specifies the subset $\chi^{-1}(\{1\})$. Therefore

$$\boxed{\mathcal{P}(X) \cong 2^X,}$$

where 2^X denotes the set of all maps from X to 2.

Ordinary set operations become pointwise Boolean operations:

$$\chi_{A \cap B} = \chi_A \chi_B, \qquad \chi_{A^c} = 1 - \chi_A,$$

and

$$\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \chi_B.$$

These formulas use arithmetic in the integers, but evaluate only to 0 or 1 on Boolean inputs.

Given $f : X \to Y$, inverse image has a particularly natural description:

$$\boxed{\chi_{f^{-1}(B)} = \chi_B \circ f.}$$

Pulling a subset back along a map is just composition of its predicate with that map. The preimage operation therefore comes from the most basic operation on functions: composition.

§1.6 Membership through a subobject classifier

A subset $A \subseteq X$ is represented by its inclusion $i : A \hookrightarrow X$. Abstractly an inclusion is a *monomorphism*: a map $m : A \to X$ with the property that $mu = mv$ forces $u = v$ for all maps u, v with a common domain. In sets this says exactly that m is injective.

Two injections can represent the same subset without their domains being literally equal. For example, a pair of distinct labels $\{\alpha, \beta\}$ could be mapped injectively to the even residues $\{\overline{0}, \overline{2}\}$ inside $\mathbb{Z}/4\mathbb{Z}$. What matters is the image and how it sits in the ambient set. Categorically we regard monomorphisms into X as equivalent when they differ by an isomorphism over X. Their equivalence classes form the collection $\mathrm{Sub}(X)$ of *subobjects*.

For sets, $\mathbf{2} = \{0, 1\}$ and the distinguished point $t : 1 \to \mathbf{2}$ with value 1 have a universal property. Each monomorphism $m : A \hookrightarrow X$ fits, up to the unique appropriate isomorphism, into a pullback square

$$\begin{array}{ccc} A & \xrightarrow{\ m\ } & X \\ \downarrow & & \downarrow{\scriptstyle \chi_A} \\ 1 & \xrightarrow[\ t\ ]{} & \mathbf{2}. \end{array}$$

Conversely every arrow $\chi : X \to \mathbf{2}$ produces a subobject by pulling back the point t. The object $\mathbf{2}$, together with t, is the *subobject classifier* of **Set**.

The two faces of a subset now meet:

$$\boxed{\mathrm{Sub}(X) \cong \mathrm{Hom}(X, \mathbf{2}) \cong \mathrm{Hom}(1, \mathbf{2}^X).}$$

The first is the classifier property; the second is currying with $T = 1$. The ordinary power set $\mathcal{P}(X)$ and the exponential $\mathbf{2}^X$ are two descriptions of the same object in **Set**.

Membership becomes a question about factorization. An element $x : 1 \to X$ belongs to the subobject represented by $m : A \hookrightarrow X$ when there is a map $a : 1 \to A$ with $ma = x$. Equivalently,

$$x \in A \quad \Longleftrightarrow \quad \chi_A x = t.$$

No membership test is required between two arbitrary mathematical objects. One first names the ambient set X, then a subobject A of X, and finally a point x of X. This is why the categorical viewpoint suits the way one actually uses sets when doing algebra or analysis.

A general elementary topos also has a subobject classifier Ω, although it need not look like $\mathbf{2}$. For sheaves on a space, truth may depend on an open region; a subobject can be locally true without having a single global Boolean value. ETCS singles out a setting in which ordinary set-like points, subsets, and classical truth values behave as expected.

§1.7 Evaluation turns Cantor's paradox into a proof

For sets X, Y, the set Y^X comes with an evaluation map

$$\mathrm{ev} : Y^X \times X \to Y, \qquad (h, x) \longmapsto h(x).$$

Suppose, toward a contradiction, that there is a surjective map

$$s : X \longrightarrow 2^X.$$

Every point $x \in X$ has been assigned a Boolean-valued function $s(x) : X \to 2$. Define the diagonal predicate

$$d(x) = 1 - \mathrm{ev}(s(x), x) = 1 - s(x)(x).$$

Because s is surjective, some $a \in X$ must satisfy $s(a) = d$. Evaluating at a gives

$$d(a) = 1 - s(a)(a) = 1 - d(a),$$

which is impossible for a value in $\{0, 1\}$. Consequently

$$\boxed{\text{there is no surjection } X \to 2^X.}$$

This is Cantor's theorem $|\mathcal{P}(X)| > |X|$. In the map-based presentation, the contradiction comes from evaluation and the operation of changing a truth value, with no need to ask what the elements of X are made of.

The same proof is familiar when written as the diagonal subset

$$D = \{x \in X : x \notin s(x)\}.$$

The categorical-looking version makes clear that the engine of the argument is a self-reference created by evaluation, not any special internal material from which sets are constructed.

For example, on $X = \{a, b\}$ suppose that a proposed list has

$$s(a) = \{b\}, \qquad s(b) = \{a, b\}.$$

The diagonal test asks whether $a \in s(a)$ and $b \in s(b)$, and reverses each answer:

	$a \in s(x)$	$b \in s(x)$	$x \in D$?
$x = a$	no	yes	yes
$x = b$	yes	yes	no

$$D = \{a\}.$$

Neither $s(a)$ nor $s(b)$ equals D. The calculation does not rely on the particular two assigned subsets: negating the answer on the diagonal guarantees that the newly constructed subset differs from every subset in the list, at its own index.

§1.8 Choice means splitting a surjection

For a surjection $p : E \to B$, a right inverse is a map $s : B \to E$ with

$$p \circ s = \mathrm{id}_B \,.$$

It chooses one element in each fiber $p^{-1}(b)$, compatibly with the ordinary definition of a function. The axiom of choice is equivalent, in the category of sets, to the assertion that every surjection has such a section.

This is a strong structural statement about maps. It fails automatically in many categories of spaces with extra structure, even when the underlying function is surjective.

Consider the continuous covering map

$$p : S^1 \to S^1, \qquad p(z) = z^2.$$

It is surjective, but there is no continuous right inverse. If s existed, the induced maps on fundamental groups would satisfy

$$p_* \circ s_* = \mathrm{id}_{\pi_1(S^1)} \,.$$

After identifying $\pi_1(S^1) \cong \mathbb{Z}$, the map p_* is multiplication by 2. Hence we would need an integer m with $2m = 1$, a contradiction.

Set-theoretically one can choose a square root of each z, but no such choice can be continuous around the circle. The example distinguishes the abstract existence of choices from their compatibility with added geometric structure.

§1.9 Natural numbers as a recursion principle

A natural numbers object N comes with a distinguished zero map $z : 1 \to N$ and successor map $s : N \to N$. Its defining property is recursion: given an object X, a point $x_0 : 1 \to X$, and a map $g : X \to X$, there is a unique $h : N \to X$ with

$$h(0) = x_0, \qquad h(n+1) = g(h(n)).$$

In ordinary sets, take $X = \mathbb{N}$, $x_0 = 0$ and $g(k) = k + 2$. The unique recursively defined map is

$$h(n) = 2n.$$

This simple property gives a structural characterization of iteration. It is a basic reason why natural numbers, finite sequences and inductive definitions fit naturally into a map-based foundation.

The same recursion principle can carry more than one number at a time. In **Set** let

$$X = \mathbb{N} \times \mathbb{N}, \quad x_0 = (0, 1), \quad g(a, b) = (b, a + b).$$

The uniquely defined arrow $h : N \to X$ begins

$$h(0) = (0, 1), \quad h(1) = (1, 1), \quad h(2) = (1, 2), \quad h(3) = (2, 3), \ldots.$$

Its two coordinates are consecutive Fibonacci numbers. One recursion on a product object generates an infinite system of linked calculations.

§1.10 The ETCS axioms in action

Lawvere's *Elementary Theory of the Category of Sets* asks for a category of sets and maps with a particularly useful package of properties. One standard modern formulation is

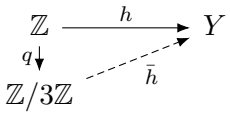
ETCS: a well-pointed elementary topos with a natural numbers object and choice.

Here an *elementary topos* has finite limits, exponential objects, and a subobject classifier. The various constructions already encountered explain what those conditions provide. Finite limits give products, equations, and fibers. Exponentials supply sets of functions. The classifier lets us form and manipulate subsets. The natural numbers object supports recursion and induction, and the choice axiom provides sections of epimorphisms.

A topos also has finite colimits, including quotients of equivalence relations. To see the connection with the elementary theory of congruences, take the relation

$$m \sim n \quad \Longleftrightarrow \quad m - n \in 3\mathbb{Z}$$

on $\mathbb{Z}$. The quotient map $q : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ identifies the pairs (m, n) with $m \sim n$. Its characteristic property is this: if $h : \mathbb{Z} \to Y$ satisfies $h(m) = h(n)$ whenever $m \sim n$, then h factors uniquely through q:



The relation is the kernel pair of q, and q is the coequalizer of its two projections. Quotients in a topos obey the same universal pattern. Hence the structures named in ETCS give a remarkably broad supply of familiar constructions without selecting particular encodings for them.

§1.11 What well-pointedness buys

ETCS has another hypothesis which deserves more attention. Its terminal object 1 is a *generator*: whenever $f, g : X \to Y$ are different arrows, some point $x : 1 \to X$ witnesses the difference,

$$f \neq g \quad \Longrightarrow \quad fx \neq gx \quad \text{for some } x : 1 \to X.$$

This is the categorical form of the ordinary principle that two functions are equal when they agree at every input. Once this property is assumed, many arrow equations may be verified by element chasing, exactly as in undergraduate mathematics.

In a general category, a map $T \to X$ is called a *generalized element* of X of shape T. If we allow all shapes T, maps can always be distinguished: choose $T = X$ and compare $f \circ \mathrm{id}_X$ with $g \circ \mathrm{id}_X$. Well-pointedness says that the single shape 1 already suffices. It is a substantive requirement, not an automatic consequence of learning the language of arrows.

The axiom of choice also has an arrow formulation:

$$\boxed{\text{every epimorphism } p : E \to B \text{ has a section } s : B \to E, \quad ps = 1_B.}$$

In **Set**, epimorphisms are surjections, so this recovers the ordinary choice axiom. ETCS asserts that the same splitting property holds in its proposed category of sets.

A useful consequence is that the internal logic of an ETCS topos is classical. In fact, Diaconescu's theorem says that the splitting of all epimorphisms in an elementary topos forces its subobject logic to be Boolean. This helps explain why the classifier of ordinary sets has precisely two truth values, while sheaf topoi can use open regions as their local truth values.

§1.12 How much set theory is hidden here?

There is a practical distinction between manipulating an already given set and forming an entirely new collection of sets. ETCS supplies subsets of each set X, quotients, products, and function sets Y^X for given objects X, Y. It can carry out a great deal of the set-based mathematics appearing in algebra, analysis, and geometry.

It does not provide an unrestricted set of all sets, and its standard axioms do not contain the full Replacement schema of ZFC. This is a real difference in proof-theoretic strength. A precise comparison is known: ETCS is *equiconsistent* with bounded Zermelo set theory with Choice, where Separation is restricted to bounded formulas and Replacement is absent. Michael Shulman's *Comparing Material and Structural Set Theories* gives a modern account. Stronger structural set theories add carefully formulated replacement or collection principles when they are needed.

A concrete distinction is useful here. Given a map $f : X \to Y$ and a subset $B \subseteq Y$, the inverse image $f^{-1}(B)$ is part of the ordinary bounded collection of subobjects of X. Replacing an entire class-indexed family of objects by a single new set is a different task: the objects to be collected may range beyond any previously named set. The extra strength required for such arguments cannot be inferred solely from the existence of Y^X.

The categorical language also makes it easier to locate which hypotheses a proof really uses. The existence of a diagonal function and the exponential 2^X is enough to formulate Cantor's argument. Choosing elements simultaneously from many fibers uses the choice axiom. Recursion uses the natural numbers object. When two arrows are compared by evaluating them on points, well-pointedness is doing the work.

§1.13 A square root with no global element

The distinctive force of well-pointedness is clearest in an example where it fails. Cover the unit circle by the map

$$p : S^1 \longrightarrow S^1, \qquad p(z) = z^2.$$

For each open arc $U \subseteq S^1$, let $\mathcal{F}(U)$ consist of continuous choices of square root:

$$\mathcal{F}(U) = \{s : U \to S^1 \text{ continuous} : s(z)^2 = z \text{ for every } z \in U\}.$$

These sets, together with restriction of functions, form a sheaf $\mathcal{F}$ on the circle. On a sufficiently short arc there are two possible continuous branches. Across overlapping arcs, a choice must agree with its restriction.

Over the entire circle there is no continuous square root. Indeed, a global $s : S^1 \to S^1$ with $s(z)^2 = z$ would give, on fundamental groups,

$$(z \mapsto z^2)_* \circ s_* = \mathrm{id}_{\mathbb{Z}},$$

forcing $2\deg(s) = 1$. Therefore

$$\boxed{\mathcal{F}(S^1) = \varnothing.}$$

A global element $1 \to \mathcal{F}$ of the sheaf topos would be exactly such a global section, so $\mathcal{F}$ has no global elements. Yet $\mathcal{F}$ is a genuine, nonempty local object: $\mathcal{F}(U)$ has elements on short arcs.

This difference prevents 1 from being a generator in the sheaf topos. Let $\underline{\mathbf{2}}$ denote the constant sheaf of two values. There are two maps

$$c_0, c_1 : \mathcal{F} \longrightarrow \underline{\mathbf{2}}$$

which assign the constant value 0 or 1 to every local section. They differ on the short arcs where sections exist. But there is no global point $x : 1 \to \mathcal{F}$ available to detect the difference, since both composites $c_0 x, c_1 x$ have no instances at all.

A single loop around the circle exchanges the two local branches. Their obstruction to gluing has a particularly concise categorical form:

$$\boxed{\mathrm{Hom}_{\mathbf{Sh}(S^1)}(1, \mathcal{F}) = \varnothing, \qquad c_0 \neq c_1 \in \mathrm{Hom}_{\mathbf{Sh}(S^1)}(\mathcal{F}, \underline{\mathbf{2}}).}$$